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HACKATHON PROJECT

TOPIC-11:ONLINE SHOPPING PRODUCT OPTIMIZER

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HACKATHON PROJECT

ShopOpt

Online Shopping Product Optimizer

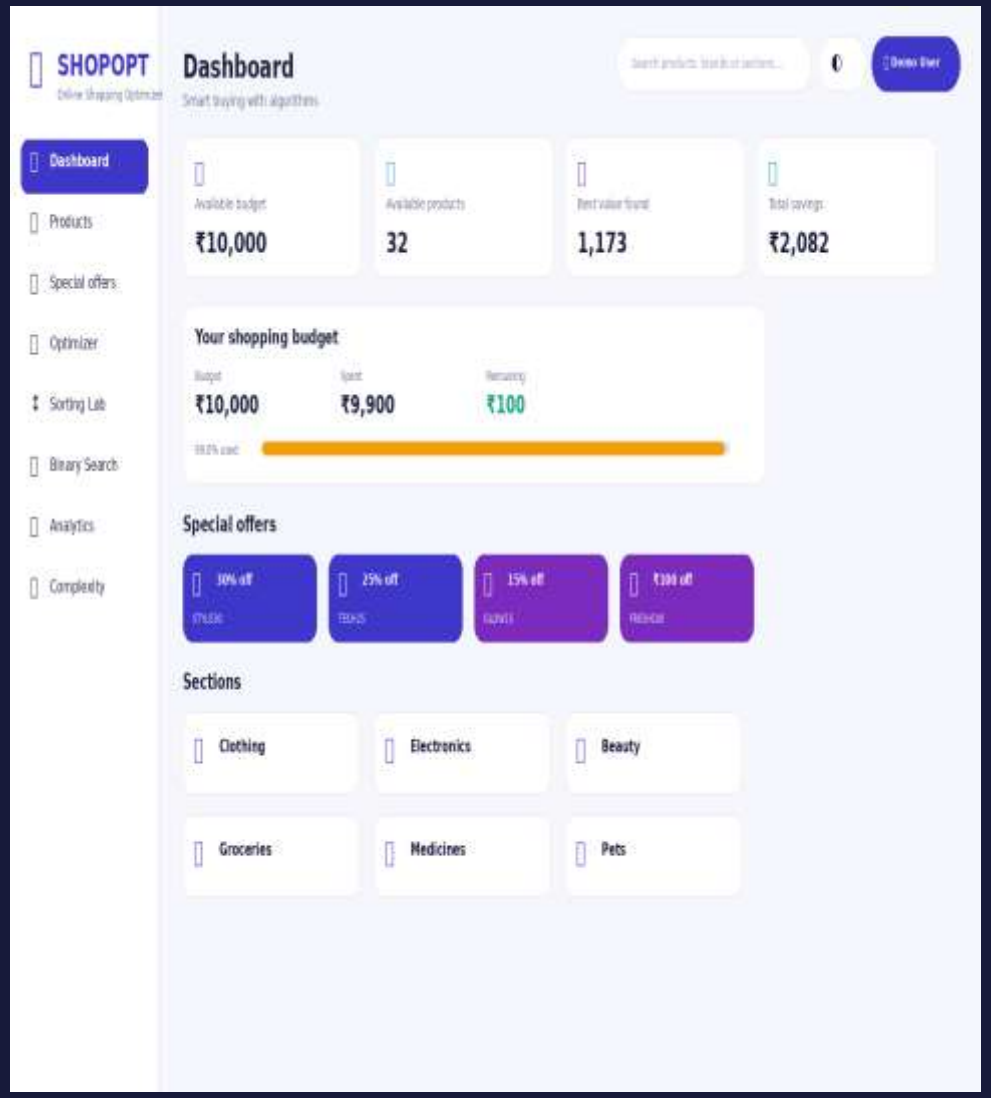
Smart buying with 0/1 Knapsack, Sorting & Binary Search

E-COMMERCE | ALGORITHM VISUALIZER | DECISION SUPPORT

Problem domain

E-commerce • Budget-aware product selection

Team members: Add team names



LIVE PROTOTYPE

02

The Challenge

Why smarter shopping needs more than a product list

The real-world problem

- Too many products make comparison difficult.
- A fixed budget can conflict with rating, utility and discounts.
- Users need a basket, not just a ranked list.
- Manual trial-and-error becomes slow as choices grow.

Core question

“What combination gives the best value without breaking the budget?”

CHOICE OVERLOAD

32 demo products across 6 sections

BUDGET CONSTRAINT

Every item consumes part of the basket

MULTI-CRITERIA

Utility • rating • discount • combined value

ALGORITHM TRANSPARENCY

Show the steps instead of hiding the logic

03

Our Innovation

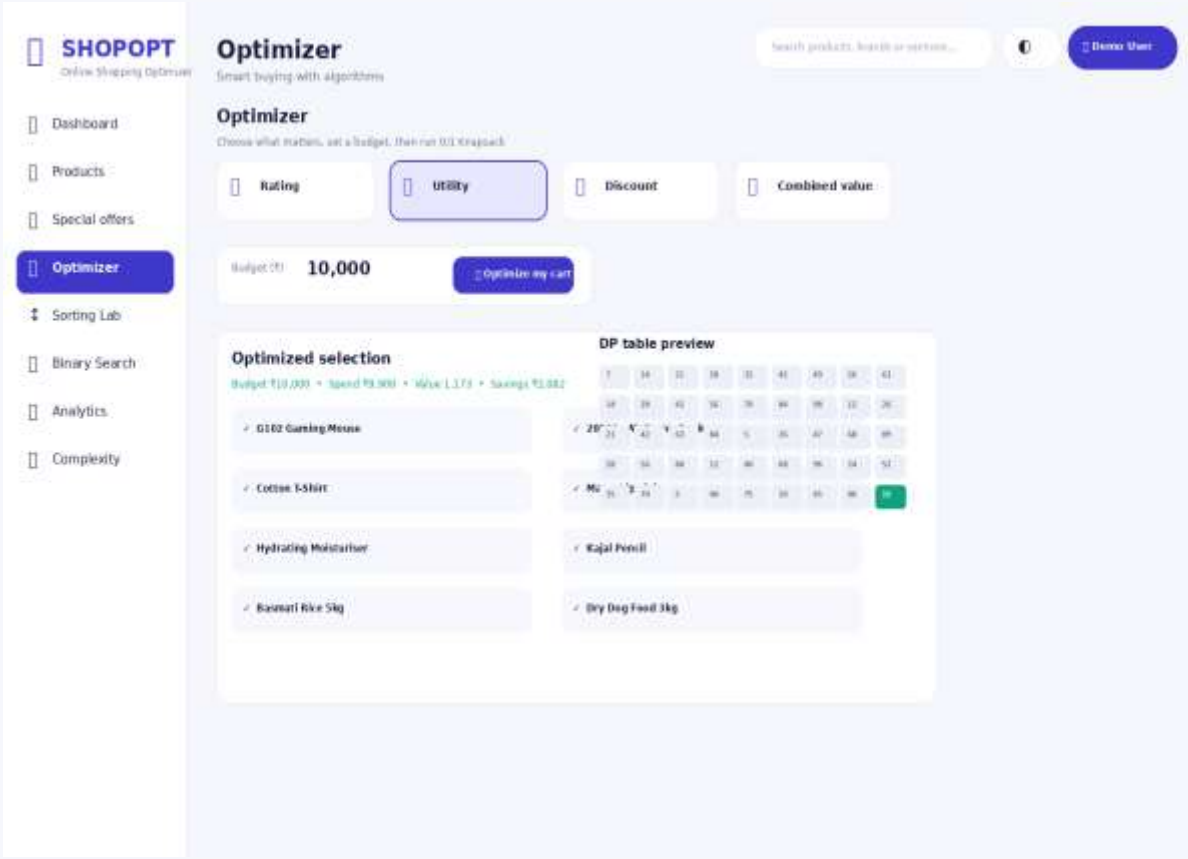
Turn a shopping decision into an interactive algorithm experience

Core idea

Optimize the whole basket — not one product at a time.

The prototype explicitly implements the algorithms so users can watch the decision process.

- 0/1 Knapsack
- Sorting Lab
- Binary Search
- Analytics



Unique approach — Same product catalogue → multiple algorithms → visible decisions

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How It Works

Simple user journey from budget to an optimized basket

1 — INPUT

Budget + products + objective

2 — PROCESS

Normalize utility / rating / discount

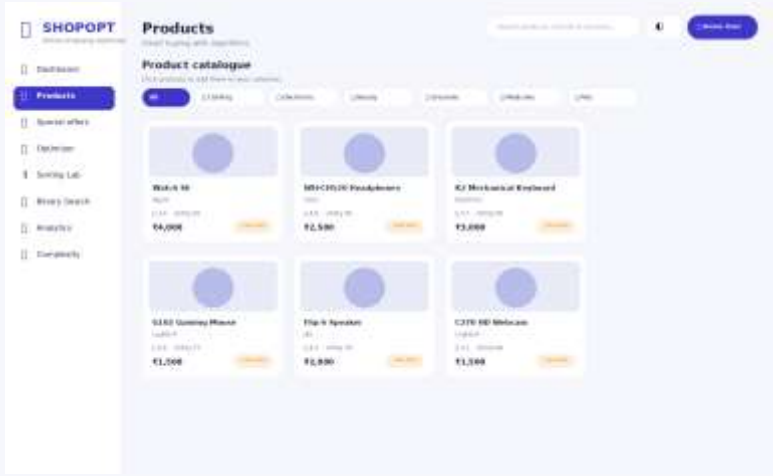
3 — ALGORITHM

Run 0/1 Knapsack DP

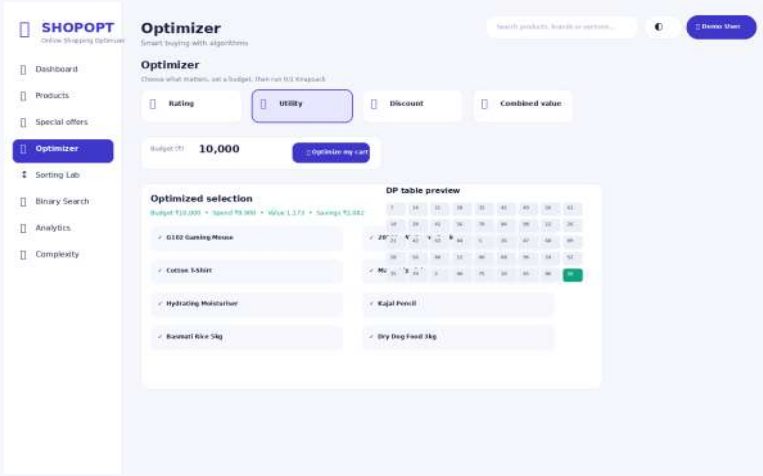
4 — OUTPUT

Best basket + spend + savings

User journey



Choose



Optimize



Measure

05

Technology & Algorithm

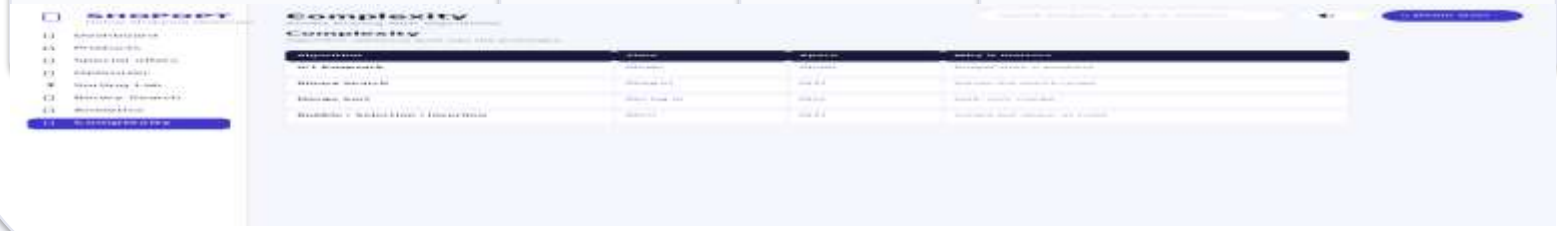
A lightweight web stack with explicit algorithm implementations

Tech stack

HTML5	UI structure
CSS3	Glass cards + responsive layout
JavaScript	UI + algorithm logic
SVG	Product illustrations
Web Speech API	Voice search
REST API	Optional backend at localhost:5000

Main algorithms

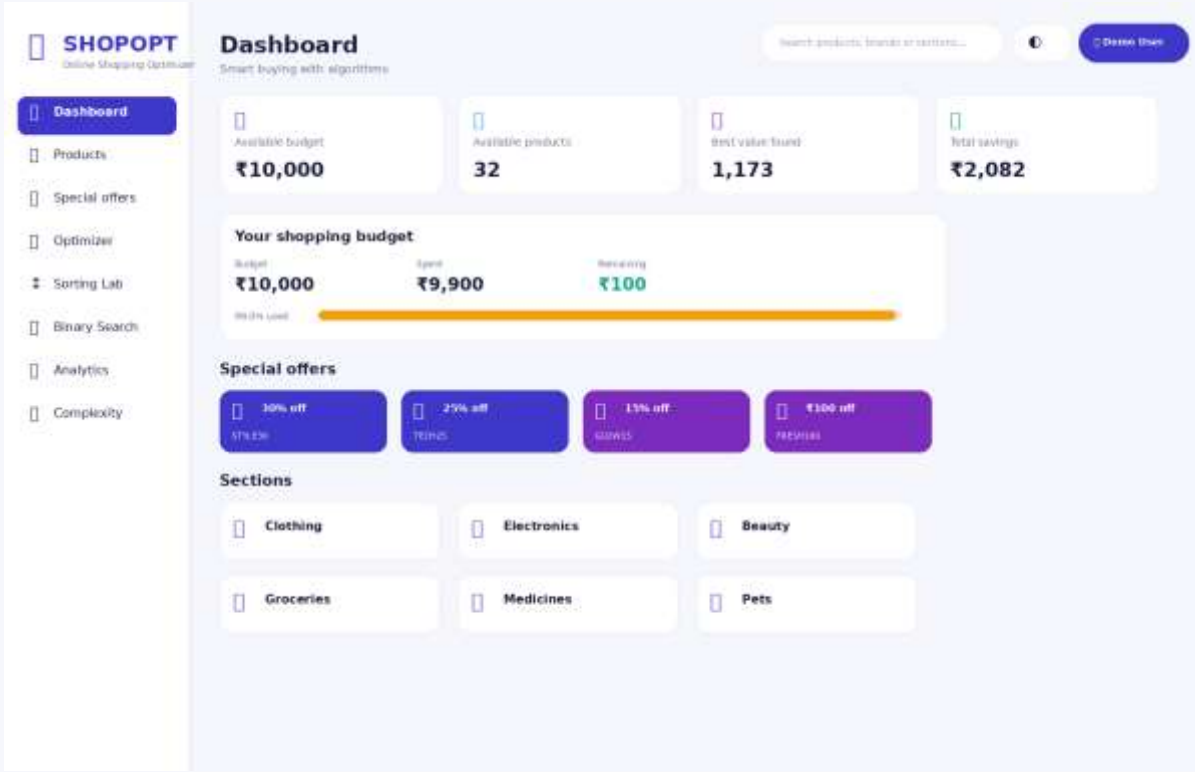
Algorithm			
0/1 Knapsack	O(nW)	O(nW)	Budget-constrained basket
Binary Search	O(log n)	O(1)	Fast lookup in sorted prices
Merge Sort	O(n log n)	O(n)	Efficient ordering



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Demo & Impact

What the prototype produces with the built-in demo catalogue



Sample result • Utility objective

₹9,900
basket spend

1,173
utility value

₹2,082 estimated discount value

32
products in demo

18
items selected

3,333
DP states

6
catalogue sections

Practical benefits — **Budget-aware decisions • Explainable algorithms • Interactive learning • Faster comparison**

07

Future Vision

From a hackathon prototype to a scalable shopping decision engine

NOW

Working prototype

Dashboard, catalogue, optimizer, sorting, binary search and analytics.

NEXT

Smarter recommendations

Live marketplace data, richer preference models, price history and personalized weights.

SCALE

Production-ready engine

API-backed catalogue, larger datasets, caching, deployment and mobile-ready UI.

Final takeaway

ShopOpt makes algorithmic shopping decisions visible, explainable and practical.

THANK YOU • Q&A